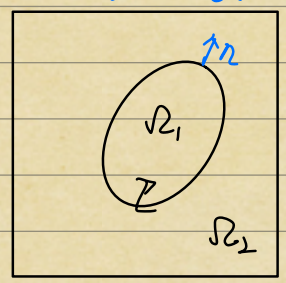
椭圆界面问题的浮面罚有限元方法 (非拟合)

$u=0$ $\partial\Omega$



$$(P)\quad \begin{cases} -\nabla\cdot(a\nabla u) = f & \text{in } \Omega_1\cup\Omega_2 \\ u|_{\partial\Omega}=0 \end{cases} \qquad [u]|_\Gamma = g_D \qquad [a\nabla u\cdot n]|_\Gamma = g_N$$

$a|_{\Omega_i} = a_i \in \mathbb{R}^+$ 高对比

记号 $A\lesssim B$ $A\le CB$ 假设 C 与 a, u 精确解 无关 $\Omega=\Omega_1\cup\Omega_2\cup\Gamma$

$$\sum_{i=1}^{2} a_i \|\nabla u\|_{H^1(\Omega_i)} \lesssim \|f\| + \|g_D\|_{H^{\frac12}(\Gamma)} + \|g_N\|_{H^{-\frac12}(\Gamma)}$$

$|au|_{H^1(\Omega_i)}$ $i=1,2$. H^1估计 也有.

$\|u\|_{L^2}$估计 依赖 $[a]$.

unfitted

§2. UIPFEM

M_h: 笛卡尔网格

$M_{h,i} = \{K\in M_h : K\cap\Omega_i\neq\phi\}$.

$M_\Gamma = \{K\in M_h : |K^\circ\cap\Gamma|\neq 0\}$

$M_{h,i}^{small} = \{K\in M_\Gamma : |K\cap M_i| \le \delta|K|\}$. $\delta\in(0,\frac12)$

2d 曲率, h 足够小, 合并单元一定能实现.

$M_{h,i}^{large} = M_{h,i}\setminus M_{h,i}^{small}$

$\forall K\in M_{h,i}^{small}$, $\exists$ 宏单元 $M_i(K)\supset K$ 和 至少一个 $K'\in M_{h,i}^{large}$ 记 $M_i(K)=M_i(K')$

其它 $K\in M_{h,i}^{large}$. 记 $M_i(K)=K$

$\forall K,K'\in M_{h,i}$ $M_i(K)=M_i(K')$ 或 $\overset{\circ}{M}_i(K)\cap\overset{\circ}{M}_i(K')=\phi$

$\cup M_i(K)\supset\Omega_i$ $\operatorname{diam}(M_i(K))\lesssim h_K$

$|M_i(K)\cap\Omega_i| \gtrsim |M_i(K)|$

定义 $T_{h,i} = \{M_i(K) : K\in M_{h,i}\}$.

$K\in T_{h,i}$ $\Omega_{h,i} = \bigcup_{K\in T_{h,i}} K$

$$V = \{v : v|_{\Omega_i} = v_i|_{\Omega_i},\ v_i\in H^1(\Omega_{h,i}),\ v_i|_K\in H^2(K)\ \forall K\in T_{h,i},\ |K\cap\Omega_i|\neq 0,\ v|_{\partial\Omega}=0\}$$

$$U_{h,i} = \{v_{h,i}\in H^1(\Omega_{h,i}) : v_{h,i}|_K\in Q_1(K),\ \forall K\in T_{h,i}\}$$

slave node

hanging

$$V_h = \{v_h : v_h|_{\Omega_i} = v_{h,i}|_{\Omega_i},\ v_{h,i}\in U_{h,i},\ v_h|_{\partial\Omega}=0\}$$

$[v] = v_1 - v_2$ $v_i = v|_{\Omega_i}$

$w_1 + w_2 = 1 \quad w_i \geq 0$

$\{v\}_w = w_1 v_1 + w_2 v_2 \qquad w_1 = \frac{a_2}{a_1+a_2} \quad w_2 = \frac{a_1}{a_1+a_2}$

$\{v\}^w = w_2 v_1 + w_1 v_2$

$a_{min} \leq \{a\}_w = \frac{2a_1a_2}{a_1+a_2} \leq 2a_{min}$ 调和权.

$[uv] = \{u\}_w [v] + [u]\{v\}^w$

$\forall v \in V$

$$(f, v) = \sum_i \int_{\Omega_i} (-\nabla\cdot(a\nabla u))v = \sum_i \left((a\nabla u, \nabla v)_{\Omega_i} - \int_{\partial\Omega_i} a\nabla u \cdot n_i v \right)$$

$$= (a\nabla u, \nabla v)_{\Omega_1\cup\Omega_2} - \int_\Gamma [a\nabla u \cdot n v]$$

$$= \cdots - \langle \{a\nabla u\cdot n\}_w [v] \rangle - \langle [a\nabla u \cdot n] \{v\}^w \rangle$$

定义 $a(u,v) = (a\nabla u, \nabla v)_{\Omega_1\cup\Omega_2} - \langle \{a\nabla u\cdot n\}_w, [v]\rangle - \beta \langle [u], \{a\nabla v\cdot n\}_w \rangle + \langle \frac{\gamma\{a\}_w}{h}[u], [v] \rangle$

$$F(v) = (f, v) + \langle g_N, \{v\}^w \rangle - \beta \langle g_D, \{a\nabla v\cdot n\}_w \rangle + \langle \frac{\gamma\{a\}_w}{h} g_D, [v] \rangle$$

则 $a(u,v) = F(v) \quad \forall v \in V$

UIPFEM：求 $u_h \in V_h$ s.t.

$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$.

2025.12.12

$\beta = 1$：SUIPFEM

$\beta = -1$：NUIPFEM

辅助结果：

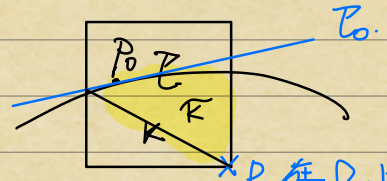
引理：存在线性延拓算子 $E_1 : H^2(\Omega_1) \longrightarrow H^2(\Omega) \cap H_0^1(\Omega)$

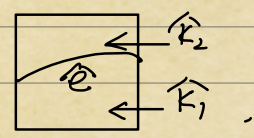
$E_2 : H^2(\Omega_2) \longrightarrow H^2(\Omega)$

$E_i(v_i)\big|_{\Omega_i} = v_i$, $\|E_i v_i\|_{H^2(\Omega)} \lesssim \|v_i\|_{2,\Omega_i}$

$\|v\|_{\partial K} \lesssim h_K^{-\frac{1}{2}} \|v\|_K + h_K^{\frac{1}{2}} \|\nabla v\|_K$.

$\|v_h\|_{\partial K} \lesssim h_K^{-\frac{1}{2}} \|v_h\|_K$.





局部迹（界面）

引理：$e = K \cap \Gamma$ $K_i^e = K \cap \Omega_i$ Γ 是 C^2 的. 则 $\exists\, h_0 > 0$

s.t. $\|v\|_e \lesssim h_K^{-\frac{1}{2}} \|v\|_{K_i^e} + \|v\|_{K_i^e}^{\frac{1}{2}} \cdot \|\nabla v\|_{K_i^e}^{\frac{1}{2}} \quad \forall v \in H^1(K)$

$\|v_h\|_e \lesssim h_K^{-\frac{1}{2}} \|v_h\|_{K_i^e} \quad \forall v_h \in V_h$. 逆估计

proof：曲线 C^2 且紧 有限开覆盖. s.t. e 一定在某个开集中，则 e 可以使用对应开集中的参数化：$e : \vec{x} = \gamma(t)$ $t \in I$ $|\gamma'(t)| \simeq 1$

平移坐标使 P 为原点 则 $e: \gamma(\xi_0)$

则 $|\gamma(t)| \lesssim h_K$

$\gamma(\xi) - \gamma(\xi_0) = \gamma'(\xi_0)(\xi-\xi_0) + O(h_K^2)$

$\gamma'(\xi) - \gamma'(\xi_0) = O(h_K)$

$\tilde{K}$ P 扫射出的扇面.

$$\tilde{K} = \{ x = t\gamma(\xi) : \xi \in I,\ t \in (0,1) \}.$$

$$(v(\gamma(\xi)))^2 = \int_0^1 \frac{d}{dt}\left(t^2 v^2(t\gamma(\xi))\right) dt. \qquad v_h \text{ 时.}$$

$$= \int_0^1 \left(2t v^2 + t^2 \cdot v \frac{d}{dt} v(t\gamma)\right) dt$$

在 P 与 $\gamma(\xi)$ 的连线上使用逆估计.

$$\lesssim \int_0^1 t\left(v^2 + |\gamma||\nabla v||v|\right) dt$$

$$dx = t|\gamma \times \gamma'| \quad dt d\xi$$

$$\|v\|_e^2 = \int_e v^2 ds = \int_I v^2(\gamma(\xi))|\gamma'(\xi)| d\xi.$$

$$\lesssim \int_0^1 \int_I t\left(v^2 + h_K |v||\nabla v|\right)|\gamma'(\xi)| d\xi dt$$

$$= \int_0^1 \int_I \left(h_K^{-1} v^2 + |v||\nabla v|\right) h_K \, t|\gamma \times \gamma'| \frac{|\gamma'|}{|\gamma \times \gamma'|} d\xi dt.$$

记: $G(\xi) = \dfrac{|\gamma(\xi) \times \gamma'(\xi)|}{|\gamma'(\xi)|}$ 则. $\|v\|_e^2 \lesssim \dfrac{h_K}{\min\limits_{\xi \in I} G(\xi)} \left(h_K^{-1}\|v\|_{\tilde{K}}^2 + \|v\|_{\tilde{K}}\|\nabla v\|_{\tilde{K}}\right)$

$$= \frac{|\gamma(\xi_0) \times \gamma'(\xi) + O(h_K^2)|}{|\gamma'(\xi_0) + O(h_K^2)|} = G(\xi_0) + O(h_K^2) \gtrsim h_K$$

$\Rightarrow G(\xi_0) \gtrsim h_K$

插值: $v \in H^2(\Omega_1 \cup \Omega_2)$

$v_i = E_i(v|_{\Omega_i})$

$v_{i,h} = I_{h,i} v_i$

$I_h v|_{\Omega_i} = v_{i,h}|_{\Omega_i}$

$\langle \{a \nabla v \cdot n\}_w \cdot [v] \rangle$

误差估计. $\|v\|^2 = (a\nabla v \cdot \nabla v)_{\Omega_1 \cup \Omega_2} + \left\langle \dfrac{\gamma \{a\}_w}{h} [v] \cdot [v] \right\rangle + \left\| \left(\dfrac{h}{\gamma \{a\}_w}\right)^{\frac{1}{2}} \{a \nabla v \cdot n\}_w \right\|_E^2$

引理: $\|v - I_h v\| \lesssim h^2 \|v\|_{H^2(\Omega_1 \cup \Omega_2)}$ $\quad \{a\}_w \le a_{min}$

$\|v - I_h v\| \lesssim h \|a^{\frac{1}{2}} v\|_{H^2(\Omega_1 \cup \Omega_2)} (1 + \gamma + \gamma^{-1})^{\frac{1}{2}}$

使用 H^2 范数而非半范, 主要是由延拓定理导致的

证: 记 $\eta_i = v_i - I_{h,i} v_i$ $\quad \eta = v - I_h v$ $\quad \eta|_{\Omega_i} = \eta_i|_{\Omega_i}$

$$\|\eta_i\|_{\Omega_{i,h}} \lesssim h^2 |v_i|_{H^2(\Omega_{i,h})} \lesssim h^2 \|v\|_{H^2(\Omega_i)}$$

$$\|\eta_i\|_{H^1(\Omega_{i,h})} \lesssim h \|v\|_{H^2(\Omega_i)}$$

$$\|\eta\| \lesssim h^2 \|v\|_{H^2(\Omega_1 \cup \Omega_2)}$$

$$\|a^{\frac12}\nabla\eta\|^2_{\Omega_1\cup\Omega_2} \lesssim \sum_i a_i\|\nabla\eta_i\|^2_{\Omega_i} \lesssim h^2\sum_i a_i\|v\|^2_{H^2(\Omega_i)}$$

$$\frac{\gamma\{a\}_w}{h}\|[\eta]\|^2_E = \frac{\gamma\{a\}_w}{h}\sum_e\sum_i\|\eta_i\|^2_e$$

$$\lesssim \frac{\gamma}{h}\{a\}_w\sum_i\sum_e\left(h^{-1}\|\eta_i\|^2_{K_i^e} + h\|\nabla\eta_i\|^2_{K_i^e}\right)$$

$$\lesssim \frac{\gamma}{h}\{a\}_w\sum_i\left(h^{-1}\|\eta_i\|^2_{\Omega_i} + h\|\nabla\eta_i\|^2_{\Omega_i}\right)$$

$\frac{(a_iw_i)^2}{\{a\}_w} \le \{a\}_w \le a_{min}$.

$$\lesssim \gamma\{a\}_w h^2\|v\|^2_{H^2(\Omega_1\cup\Omega_2)}$$

$$\frac{h}{\gamma\{a\}_w}\|\{a\nabla\eta\cdot n\}_w\|^2_E \lesssim \frac{h}{\gamma\{a\}_w}\sum_i\sum_e\|a_iw_i\nabla\eta_i\|^2_e$$

$$\lesssim \frac{h}{\gamma\{a\}_w}\sum_i (a_iw_i)^2\left(h^{-1}\|\nabla\eta_i\|^2_{\Omega_i} + h|\eta_i|^2_{H^2(\Omega_1\cup\Omega_2)}\right)$$

$$\lesssim \frac{h}{\gamma\{a\}_w}\sum_i (a_iw_i)^2 h\|v\|^2_{H^2(\Omega_i)}$$

2025.12.1

$(\pi_m v, q_e)_{H^m} = (v, q_e)_{H^m} \quad \forall q_e\in P_{m-1}$

$\inf_{q_e\in P_{m-1}}\|u-q_e\|_{H^m} = \|v-\pi_m v\|_{H^m} \lesssim |v|_{H^m}$ Deny-Lions 引理

$\inf_{q_e\in P_1}\|v-q_e\|_{H^2} \lesssim |v|_{H^2}$

$q_e\in P_1(\Omega_1\cup\Omega_2)$

$\|\|v-q_e - I_h(v-q_e)\|\| \lesssim (1+\gamma+\gamma^{-1})^{\frac12} h\|a^{\frac12}(v-q_e)\|_{H^2(\Omega_1\cup\Omega_2)}$

$I_h(v-q_e)$ ⇑ 有限元空间

引理: $v\in H^2(\Omega_1\cup\Omega_2)$ $v|_{\partial\Omega}=0$ 则 $\inf_{v_h\in V_h}\|\|v-v_h\|\| \lesssim (1+\gamma+\gamma^{-1})^{\frac12} h|a^{\frac12}v|_{H^2(\Omega_1\cup\Omega_2)}$

SUIPFEM: $\beta=1$ 误差估计

$$\|\|v\|\|^2 = (a\nabla v\cdot\nabla v)_{\Omega_1\cup\Omega_2} + \left\langle\frac{\gamma\{a\}_w}{h}[v],[v]\right\rangle + \left\|\left(\frac{h}{\gamma\{a\}_w}\right)^{\frac12}\{a\nabla v\cdot n\}_w\right\|^2_E.$$

引理 (i) $|a(u,v)| \lesssim \|\|u\|\|\,\|\|v\|\|$

(ii) $\exists\,\gamma_0>0$, $\gamma\ge\gamma_0$ 时, $a(v_h,v_h)\ge\frac12\|\|v_h\|\|^2 \quad\forall v_h\in V_h$.

证: (ii) $a(v_h,v_h) = \|\|v_h\|\|^2 - 2\left\langle\{a\nabla v_h\cdot n\}_w,[v_h]\right\rangle - \left\|\frac{h}{\gamma\{a\}_w}\{a\nabla v_h\cdot n\}_w\right\|^2_E$

$$\ge \|\|v_h\|\|^2 - \frac12\left\|\left(\frac{\gamma\{a\}_w}{h}\right)^{\frac12}[v_h]\right\|^2_E - \underbrace{3\left\|\left(\frac{h}{\gamma\{a\}_w}\right)^{\frac12}\{a\nabla v_h\cdot n\}_w\right\|^2_E}_{(I)}$$

$$(I) \lesssim \frac{h}{\gamma\{a\}_w}\sum_i \|a_i w_i \nabla u_{h,i}\|_E^2 \lesssim \frac{h\{a\}_w^2}{\gamma\{a\}_w}\sum_i h^{-1}\|\nabla v_{h,i}\|_{\Omega_i}^2$$

$$\lesssim \frac{\{a\}_w}{\gamma}\sum_i \|\nabla v_{h,i}\|_{\Omega_i}^2$$

$$\lesssim \frac{1}{\gamma}\sum_i \|a_i^{\frac12}\nabla v_{h,i}\|^2$$

$$\le \frac{c}{\gamma}(a\nabla v_h,\nabla v_h)_{\Omega_1\cup\Omega_2}$$

(Cea)引理 $\gamma\ge\gamma_0 \Rightarrow \quad |||u-u_h||| \lesssim \inf_{v_h\in V_h} |||u-v_h|||$

proof: $\forall v_h\in V_h$

$$|||u_h-v_h|||^2 \lesssim a(u_h-v_h,u_h-v_h)$$

$$= a(u-v_h,u_h-v_h)$$
$$\le |||u-v_h|||\,|||u_h-v_h|||$$

$$|||u-u_h||| \le |||u-v_h||| + |||u_h-v_h|||$$

$$\lesssim |||u-v_h|||$$

定理(SUIPFEM, H^1) 设 $\gamma\ge\gamma_0$. 则 $|||u-u_h||| \lesssim (1+\gamma)^{\frac12} h\,|a^{\frac12}u|_{H^1(\Omega_1\cup\Omega_2)}$

[注]: $\|a^{\frac12}\nabla(u-u_h)\|_{\Omega_1\cup\Omega_2} \lesssim (1+\gamma)^{\frac12}h\,|a^{\frac12}u|_{H^2(\Omega_1\cup\Omega_2)}$

可以证明 $\|a\nabla(u-u_h)\| \lesssim (1+\gamma)^{\frac12}h\,|au|$

相对误差可能可以去掉高对比系数影响.

$$\|[u-u_h]\|_E \lesssim h^{\frac12}a_{min}^{-\frac12}|a^{\frac12}u|_{H^2(\Omega_1\cup\Omega_2)}$$

2025.12.19

L^2 误差估计.

(DP)
$$\begin{cases} -\nabla\cdot(a\nabla\varphi) = u-u_h, & \Omega_1\cup\Omega_2 \\ [\varphi]=0 \quad [a\nabla\varphi\cdot n]=0 & \text{on } E \\ \varphi|_{\partial\Omega}=0 \end{cases}$$

$$a(\varphi,v) = (u-u_h,v)$$

$$j=1,2. \quad |a_i\varphi|_{H^s(\Omega_i)} \lesssim \|u-u_h\|$$

$$\|u-u_h\|^2 = a(\varphi,u-u_h) = a(u-u_h,\varphi) = a(u-u_h,\varphi-\varphi_h) \lesssim |||u-u_h|||\,|||\varphi-\varphi_h|||.$$

$$\Rightarrow \|u-u_h\|^2 \lesssim \inf_{\varphi_h\in V_h} |||\varphi-\varphi_h|||\,|||u-u_h|||$$

$\lesssim \|u-u_h\|\ \gamma^{\frac{1}{2}} h\ |a^{\frac{1}{2}}\varphi|_{H^2(\Omega_1\cup\Omega_2)}$ 非黑面 L^∞ 估计 ⇒ 黑面 L^∞ 估计

$\lesssim \cdots \quad \gamma^{\frac{1}{2}} h\ a_{min}^{-\frac{1}{2}} |a\varphi|$

$\lesssim$ ——————— $\|u-u_h\|$

定理：($\beta=1$) $\gamma \geq \gamma_0 \Rightarrow \|u-u_h\| \lesssim a_{min}^{-\frac{1}{2}}\ \gamma h^2\ |a^{\frac{1}{2}}u|_{H^2(\Omega_1\cup\Omega_2)}$

NUIPFEM. $\beta=-1$

引理 (i) $|a(u,v)| \lesssim \|u\|\|v\|$

(ii) $a(v_h,v_h) \gtrsim \frac{\gamma}{1+\gamma}\|v_h\|$ $\forall v_h \in V_h$.

证：$a(v_h,v_h) = \|v\|^2 - \|\left(\frac{h}{\gamma\{a\}_w}\right)^{\frac{1}{2}}\{a\nabla v_h\cdot n\}_w\|_E^2 \lesssim \frac{1}{\gamma}\|a^{\frac{1}{2}}\nabla v_h\|^2_{\Omega_1\cup\Omega_2} \lesssim \frac{1}{\gamma}a(v_h,v_h)$

引理 (Céa) 设 $\gamma \gtrsim 1$ 则. $\|u-u_h\| \lesssim \inf_{v_h\in V_h}\|u-v_h\|$

定理：($\beta=-1$) $\gamma \gtrsim 1$ 则 $\|u-u_h\| \lesssim \gamma^{\frac{1}{2}} h\ |a^{\frac{1}{2}}u|_{H^2(\Omega_1\cup\Omega_2)}$

L^2 估计： $\|u-u_h\|^2 = a(\varphi, u-u_h)$

$= a(u-u_h, \varphi) - 2\langle [u-u_h], \{a\nabla\varphi\cdot n\}_w\rangle$

$= a(u-u_h, \varphi-\varphi_h) - \cdots$

$\lesssim \|u-u_h\|\|\varphi-\varphi_h\| + \|[u-u_h]\|_E \underline{\|\{a\nabla\varphi\cdot n\}_w\|_E}$.

②

② $\lesssim \sum_i \|a_i w_i \nabla\varphi_i\|_E \lesssim \sum_i a_i w_i \|\nabla\varphi_i\|_{H^1(\Omega_i)} \lesssim \|u-u_h\|$

$\|u-u_h\| \lesssim \gamma^{\frac{1}{2}} a_{min}^{-\frac{1}{2}}\ h\|u-u_h\| + \|[u-u_h]\|_E$.

定理：($\beta=-1$) $\gamma \gtrsim 1$. $\|u-u_h\| \lesssim a_{min}^{\frac{1}{2}}\ (\gamma h^2 + h^{\frac{3}{2}})\ |a^{\frac{1}{2}}u|_{H^2(\Omega_1\cup\Omega_2)}$

数值不存在 掉阶. 高次元负模估计.

条件数 ($\beta=1$)

$AU=F$ $\quad cond(A) = \frac{\lambda_{max}}{\lambda_{min}}$. $\quad \frac{a_{max}}{a_{min}} h^{-2}$ 拟合 FEM.

$V_h = span\{\phi_1 \cdots \phi_N\}$. $\forall v_h \in V_h$ $v_h = v_1\phi_1 + \cdots + v_N\phi_N$

$V = (v_1, \cdots, v_N)^T$ SUIPFEM: $AU=F$ $\quad A = (a(\phi_j,\phi_i))_{N\times N}$.

$a(v_h,v_h) = v^T A v$, $M = (\phi_j,\phi_i)_{N\times N}$ $\quad (v_h,v_h) = v^T M v$

$A_h : V_h \to V_h$. $\quad (A_h u_h, v_h) = a(u_h, v_h)$

$$\frac{v^TAv}{v^Tv} = \frac{a(v_h, v_h)}{(v_h, v_h)} \quad \frac{v^TMv}{v^Tv} = \frac{(A_hv_h, v_h), v^TMv}{(v_h, v_h)\ v^Tv}$$

定义三个条件数: $cond(A) = \frac{\lambda_{max}^A}{\lambda_{min}^A}$, $cond(M) = \frac{\lambda_{max}^M}{\lambda_{min}^M}$

$$cond(A_h) = \frac{\lambda_{max}^{A_h}}{\lambda_{min}^{A_h}}$$

$$cond(A) \leq cond(A_h) \cdot cond(M)$$

估计 $cond(M)$

引理: $k \in T_{h,i}$, $|K \cap \Omega| \neq 0$ 则 $\forall q_k \in Q_1(K)$ $\|q_k\|_K \lesssim \|q_k\|_{K \cap \Omega_i}$

$v_h \in V_h$, $v_h|_{\Omega_i} = v_{h,i}|_{\Omega_i}$

$v_{h,1} \in V_{h,1}$ $v_{h,2} \in \{ w_{h,2} \in V_{h,2} \mid w_{h,2}|_{\partial\Omega} = 0 \} = V_{h,2}^0$

$V_{h,1} = span(\phi_{1,1} \dots \phi_{1,N_1})$ $V_{h,2}^0 = (\phi_{2,1} \dots \phi_{2,N_2})$

$v_{h,i} = \sum_{j=1}^{N_i} v_{i,j} \phi_{i,j}$ $v^i = (v_{i,1} \dots v_{i,N_i})^T$

$v^Tv = (v^1)^Tv^1 + (v^2)^Tv^2$

引理: $(v_h, v_h) \eqsim \sum_{i=1}^{2} (v_{h,i}, v_{h,i})_{\Omega_{h,i}}$

$\|v_{h,i}\|_{\Omega_{h,i}}^2 = \sum_{K \in T_{h,i}} \|v_{h,i}\|_K^2 \eqsim h^d (v_i)^T(v_i)$ 不考虑对 P 的依赖

$v^TMv = \|v_h\|^2 \eqsim h^d \sum_i (v_i)^T(v_i) \eqsim h^d\, v^Tv$

$\Rightarrow$ $\frac{v^TMv}{v^Tv} \eqsim h^d$ $\lambda_{max}^M \eqsim \lambda_{min}^M$ $\Rightarrow$ $cond(M) = O(1)$